

Project 2

CS 261

Save your program for this project as `<emailID>_project2.py` where `<emailID>` is the part of your Hampden-Sydney e-mail address before the `@` symbol. When you are finished, e-mail your program to `blins@hsc.edu`. Your solution is due by noon on Friday, October 2.

Caesar Cipher

The Caesar cipher is one of the oldest and simplest methods for encrypting text. You pick a shift amount and then you shift every letter in the message by that amount. For example, if the shift is 2, then a's become c's, b's become d's and so on, until y and z wrap around to become a and b, respectively. A message like:

about to get hazy

would become:

cdqvv vq igv jcba.

For this project, you will write three functions related to the Caesar cipher. You may also include any additional functions that are helpful to construct the three functions described below. You should not import any libraries for this project.

1. Write a function `letter2number` that converts lower case letters (a, b, c, ..., z) to the numbers (0, 1, 2, ..., 25). For example `letter2number("c")` should return 2.
2. Write a function `number2letter` that converts a number to the corresponding (lower case) letter of the alphabet. For example, `number2letter(1)` should return "b".
3. Write a function called `caesar_shift(message, shift)` that applies a Caesar shift to any string. You can assume that the message will not have any upper case letters, and you only need to shift letters, not any other symbols.
4. The Caesar cipher is not a good way to encrypt a message because it is very easy to decrypt. In fact, you can often figure out the shift by finding the most common letter in the encrypted text. Since "e" is the most common letter in most English texts, there is a good chance that the shift is just the amount needed for "e" to become the most common letter in your encrypted message. Write a function called `letter_frequency(message)` that returns a list. The value of the output list at index number *i* should be the number of times the *i*-th letter of the alphabet occurs in the message. So the first entry of the output is the number of a's, then b's, and so on.

Here is a link with some tips on how to get started:

<https://bclins.github.io/fall26/coms261/project2tips.pdf>